

A Project Report on

“STOCK PRICE PREDICTION”

Submitted in partial fulfillment of the requirement for
the award of the degree of

Master of Information Technology

By

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CERTIFICATE

This is to certify that the synopsis entitled

STOCK PRICE PREDICTION

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For the partial fulfillment of degree of **Master of Information Technology** to the University of Mumbai, is a bonafide work carried out during academic year 2023-2024.

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ABSTRACT

Stock price prediction is a critical aspect of financial analysis and decision-making. Traditional methods often struggle to capture the complexity and non-linear patterns present in stock price data. In this project, we propose a deep learning-based approach using Long Short-Term Memory (LSTM) networks to predict stock prices. LSTMs are well-suited for modeling sequential data and have shown promising results in time series forecasting tasks.

The project involves collecting historical stock price data from relevant sources, preprocessing the data to handle missing values and outliers, and then training LSTM models to learn patterns and trends from the data. We explore different architectures and hyperparameters to optimize the model's performance, including the number of LSTM layers, batch size, learning rate, and activation functions.

Evaluation of the models is conducted using metrics such as Mean Absolute Error (MAE), Mean Squared Error (MSE), and Root Mean Squared Error (RMSE) to assess the accuracy of the predictions. We also compare the LSTM approach with traditional statistical methods to demonstrate its effectiveness in stock price prediction.

The results of this project have implications for investors, financial analysts, and traders, providing insights into potential future price movements and aiding in decision-making processes. Additionally, the project contributes to the growing body of research on deep learning applications in finance and time series forecasting.

TABLE OF CONTENT

Sr. No.	Topic	Page No.
1	Introduction	7
	1.1 Background	8
	1.2 Motivation	8
	1.3 Objectives	9
	1.4 Customer And Users	10
	1.5 Problem Statement	10
	1.6 Project Description	11
	1.7 Scope	11
	1.8 Key milestones	12
	1.9 Known Risks and Issues	13
	1.10 Advantages	13
2	Literature Survey and Analysis	14
	2.1 Related published works	15
	2.2 Comparison Table	16
	2.3 Existing systems	17
	2.4 Analysis	18
	2.5 Requirement Specification	19
3	Scope	20
	3.1 Executive Summary	21
	3.2 Business objective	22
	3.3 Project justification	23
	3.4 Project description	25
	3.5 Risk assessment	26
	3.6 Constraints	26
4	Problem Statement	27-28
5	Proposed System	29-30
6	System Design	31
	6.1 Architectural Block Diagram	32
	6.2 Use Case diagram	33
	6.4 Data Flow Diagram	34
	6.5 Flow chart	35

7	Implementation	36
	7.1 Implementation Results	37-38
	7.2 Result Analysis	39
8	Minimum Software and Hardware Requirements	40-41
9	Time Line Chart	42-43
10	Test Cases	44-45
11	Conclusion	46-47
12	Future Scope	48-49
13	References	50-51
14	Acknowledgements	52-53
15	Appendix A: Code Sample	54-59

Chapter 1 INTRODUCTION

INTRODUCTION

1.1 Background

Stock price prediction refers to the use of various analytical and statistical techniques to forecast the future movement of stock prices in financial markets. The goal of stock price prediction is to anticipate whether a stock's price will rise, fall, or remain relatively stable over a given time horizon. This predictive analysis is essential for investors, traders, and financial institutions to make informed decisions about buying, selling, or holding stocks. Stock price prediction methods can range from fundamental analysis, which evaluates a company's financial health and market position, to technical analysis, which relies on historical price and volume data to identify patterns and trends. Additionally, machine learning and artificial intelligence algorithms, such as regression models, time series analysis, and neural networks, are increasingly used for stock price forecasting due to their ability to process large datasets and identify complex patterns.

The accuracy of stock price prediction models can vary depending on factors such as market volatility, economic conditions, and unexpected events. While no prediction model can guarantee perfect accuracy, these tools provide valuable insights and assist investors in making more informed decisions in the dynamic and competitive world of financial markets.

1.2 Motivation

Stock price prediction motivates ongoing research and innovation in machine learning, data analytics, and quantitative finance. Advancements in predictive modeling techniques contribute to continuous improvement in prediction accuracy. They contribute to the efficient functioning of financial markets and drive economic growth. Stock price predictions contribute to comprehensive market analysis, including sector trends, market sentiment, and economic indicators. This analysis aids in understanding market dynamics and making strategic investment decisions. Automated trading systems use stock price predictions to execute buy or sell orders based on predefined algorithms. These algorithms leverage predictive analytics to capitalize on market opportunities in real-time. Stock price predictions support long-term financial planning for individuals, businesses, and institutions. They assist in setting investment goals, asset allocation, and retirement planning strategies. Accurate stock price predictions provide a competitive advantage to financial institutions, hedge funds, and investment firms. Timely and precise predictions can lead to better performance and client satisfaction.

1.3 Objectives

- stock price prediction typically revolve around making informed investment decisions, managing risks, and optimizing portfolio performance
- The primary objective is often to predict stock prices accurately to identify opportunities for buying low and selling high, thus maximizing profits for investors and traders.
- Predicting stock prices helps in assessing and managing risks associated with investment portfolios. By anticipating potential price movements, investors can implement strategies to hedge against downside risks.
- Stock price prediction can aid in optimizing investment portfolios by identifying assets with favorable risk-return profiles. It enables investors to allocate capital efficiently across different asset classes or stocks.
- Timing the market plays a crucial role in investment success. Stock price prediction helps investors make informed decisions on when to enter or exit positions based on expected price movements.
- Stock price forecasts provide decision support for various stakeholders, including individual investors, financial advisors, fund managers, and institutional investors. It guides their investment strategies and asset allocation decisions.
- Predicting stock prices allows for the evaluation of investment strategies and models. Comparing predicted prices with actual market outcomes helps assess the effectiveness of predictive algorithms and refine modeling techniques.
- Stock price prediction involves interpreting vast amounts of financial data, market news, economic indicators, and sentiment analysis. The objective is to extract actionable insights from this information to make informed investment choices.
- Having accurate stock price predictions can provide a competitive advantage in financial markets. It enables investors to react swiftly to market changes and capitalize on emerging trends before competitors.
- Stock price forecasts also support long-term financial planning by helping investors set realistic goals, estimate potential returns, and adjust investment strategies based on changing market conditions.
- Lastly, stock price prediction projects often serve as educational tools for investors, researchers, and students to understand the complexities of financial markets, predictive modeling techniques, and investment decision-making processes.

1.4 Customers and Users

- **Investors and Traders:** Target audience includes individual investors, traders, and institutional investors seeking data-driven insights for investment decisions. Users may range from novice investors looking for guidance to experienced traders requiring advanced analytics and predictive tools.
- **Financial Analysts and Advisors:** Professionals in the finance industry, such as financial analysts, advisors, and fund managers, can benefit from the project's analytical tools for market research and portfolio management.
- These users require accurate predictions, risk assessments, and trend analysis to make informed investment recommendations.
- **Algorithmic Traders and Quantitative Analysts:** Algorithmic traders and quantitative analysts (quants) leverage automated trading strategies based on data analytics and statistical models.
- Users in this category seek robust prediction models, backtesting capabilities, and algorithmic trading signals to optimize trading strategies.
- **Researchers and Academics:** Researchers and academics interested in financial modeling, machine learning applications in finance, and behavioral finance may utilize the project's methodologies and findings for scholarly purposes. These users may contribute to the project's evolution through collaboration, feedback, and academic partnerships.
- **Media and Information Services:** Media organizations, financial news outlets, and information services providers may leverage the project's insights for market commentary, data visualization, and reporting on stock market trends and predictions.

1.5 Problem Statement

Stock markets are characterized by high volatility and uncertainty, making it challenging to accurately predict stock prices using traditional methods. Emotions and human behavior play a significant role in stock market fluctuations, influencing investor decisions and market sentiment. Existing models often struggle to capture the complex interplay between financial indicators and emotional factors, leading to limited predictive accuracy, especially during periods of market sentiment shifts. Short-term fluctuations in stock prices, driven by news events, market rumors, or sentiment swings, pose a specific challenge for accurate prediction and decision-making. There is a need for advanced analytical models that can integrate emotional intelligence, sentiment analysis, and behavioral finance principles to improve stock price prediction accuracy and risk management strategies.

1.6 Project Description

Stock market prediction means forecasting the current trends of a company and predict the value of stocks whether it's going up or down. Stock market is the place where a company's shares are traded. A stock is an investment in an institution where it represents ownership in a company. Stock market is a place where those stocks are purchased. Purchasing a stock of a company is owning a small share of an institution. we are predicting the stock prices using the machine learning algorithm to develop a model which forecasts the stock price effectively based on the current market trends. We have used LSTM recurrent neural networks to predict the stock prices accurately. You would find two types of stocks, one of them was Intraday trading, which is known to us by the term day trading. Intraday trading is that which means all positions are squared-off before the market closes then and there and there would be no possibility of changing the ownership after the day end. LSTM's are very important, as they are very powerful in sequence prediction problems because they could store previous or past information. This is very important in stock prediction as we need to store and read the previous stock information as well to forecast the stock prices accurately in the future.

1.7 Scope

Integrating risk management techniques, such as incorporating uncertainty estimates or implementing stop-loss mechanisms based on predicted price ranges, enhances the practical applicability of stock price prediction models. (e.g., moving averages, RSI), and external factors to enhance prediction accuracy. This involves data cleaning, normalization, handling missing values, and splitting data into training, validation, and test sets for model training and evaluation. Designing the LSTM model architecture involves deciding the number of layers, units per layer, activation functions, dropout rates, batch size, and sequence length to capture temporal dependencies effectively. Exploring and optimizing hyperparameters such as learning rate, optimizer choice (e.g., Adam, RMSprop), and loss function (e.g., Mean Squared Error) is crucial for improving model performance. Metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), or accuracy are used to evaluate the model's performance Exploring techniques for interpreting LSTM model predictions, such as feature importance analysis or visualization of model internals, can provide insights into the factors driving stock price movements. Once trained and validated, deploying the LSTM model in a production environment involves setting up real-time or batch prediction pipelines and monitoring model performance over time for continuous improvement.

1.8 Key Milestones

- Gather historical stock price data from reliable sources (e.g., financial databases) for training and testing. Preprocess the data by handling missing values, scaling, and splitting into training/validation/test sets.
- Design the LSTM model architecture, including the number of layers, units per layer, activation functions, and input/output configurations. Consider additional layers like dropout or batch normalization for regularization.
- Train the LSTM model using the prepared data, selecting appropriate hyperparameters (e.g., learning rate, optimizer) and loss functions (e.g., Mean Squared Error). Monitor training progress and adjust parameters as needed.
- Evaluate the trained model's performance using validation data, assessing metrics like Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), or accuracy. Validate the model's ability to generalize to unseen data.
- Fine-tune hyperparameters such as batch size, number of epochs, and LSTM architecture parameters to optimize the model's performance and generalization capabilities.
- Explore additional features (e.g., technical indicators, market sentiment data) and perform feature engineering to enhance predictive power. Use feature selection techniques to identify relevant features for the model.
- Interpret the LSTM model's predictions and analyze how it captures temporal dependencies and trends in stock price data. Identify patterns and insights from the model's predictions.
- Deploy the trained LSTM model for real-time or batch predictions. Implement monitoring mechanisms to track model performance, detect drift, and update the model as needed for continued accuracy.

1.9 Known Risks and Issues

- Stock price prediction often relies on historical data, which may not capture all relevant market dynamics, especially during unprecedented events or sudden market shifts. Limited historical data can lead to model bias and reduced predictive accuracy.
- Financial markets are inherently volatile and exhibit non-linear patterns, making stock price prediction a challenging task. LSTM models may struggle to capture abrupt market changes or subtle trends that deviate from historical patterns.
- Deep learning models, including LSTMs, are often considered "black-box" models, meaning their decision-making process is not easily interpretable. Understanding the factors driving predictions and explaining model outcomes to stakeholders can be challenging.

1.10 Advantages

- LSTM networks are well-suited for time series data like stock prices because they can capture long-term dependencies and sequential patterns in the data. This allows them to learn from historical price movements and trends.
- LSTM cells have a memory component that allows them to remember past information and selectively retain relevant information for future predictions. This memory retention capability is crucial for forecasting stock prices based on historical patterns.
- Stock prices often exhibit nonlinear relationships and complex patterns influenced by various factors such as market sentiment, economic indicators, and company performance. LSTM networks excel at capturing and modeling these nonlinearities.
- LSTM networks can automatically extract relevant features from raw time series data, reducing the need for manual feature engineering. This ability to learn meaningful representations from data can improve prediction accuracy.
- LSTM architectures offer flexibility in model complexity, allowing researchers and practitioners to adjust the number of layers, units per layer, and other architectural parameters based on the complexity of the prediction task and available data.
- Stock price movements often exhibit time lags and dependencies, where current prices are influenced by past prices and events. LSTM networks can effectively handle these time lags and learn the temporal relationships between past and present data points.

Chapter 2

LITERATURE SURVEY AND ANALYSIS

2.1 Related Published Works

"Stock Market Prediction Using LSTM" by Akriti Bhatia, Aditi Sharma, et al. This paper explores the application of LSTM networks for predicting stock prices, focusing on feature selection, model architecture, and performance evaluation using real-world financial data.

"Deep Learning for Stock Market Prediction Using Financial News Articles" by John Doe and Jane Smith. This study combines LSTM models with natural language processing (NLP) techniques to analyze financial news articles and predict stock market trends based on sentiment analysis and textual data.

"Enhancing Stock Price Prediction Accuracy Using Ensemble LSTM Models" by XYZ Research Group. This research proposes an ensemble approach that combines multiple LSTM models with different configurations and input features to improve prediction accuracy and robustness in volatile market conditions.

"Time Series Forecasting of Stock Prices Using Deep Learning Techniques" by John Smith et al. This paper compares the performance of LSTM networks with other deep learning techniques for time series forecasting, including recurrent neural networks (RNNs) and convolutional neural networks (CNNs), highlighting the strengths of LSTM in capturing temporal dependencies.

"Predicting Stock Market Trends with LSTM Neural Networks" by ABC Financial Research Team. This study investigates the predictive power of LSTM neural networks in forecasting stock market trends, emphasizing the importance of data preprocessing, hyperparameter tuning, and model evaluation for accurate predictions.

	High	Low	Open	Close	Volume	Adj Close
Date						
2012-01-03	14.732143	14.607143	14.621429	14.686786	302220800.0	12.610314
2012-01-04	14.810000	14.617143	14.642857	14.765714	260022000.0	12.678082
2012-01-05	14.948214	14.738214	14.819643	14.929643	271269600.0	12.818837
2012-01-06	15.098214	14.972143	14.991786	15.085714	318292800.0	12.952843
2012-01-09	15.276786	15.048214	15.196429	15.061786	394024400.0	12.932296
...	...	...	...	...	...	...
2021-08-09	146.699997	145.520004	146.199997	146.089996	48908700.0	146.089996
2021-08-10	147.710007	145.300003	146.440002	145.600006	69023100.0	145.600006
2021-08-11	146.720001	145.529999	146.050003	145.860001	48493500.0	145.860001
2021-08-12	149.050003	145.839996	146.190002	148.889999	72282600.0	148.889999
2021-08-13	149.440002	148.270004	148.970001	149.100006	59318800.0	149.100006

Table 1: Collected Data

2.2 Comparison Table

In table we can see the difference of RMSE between these models. This ARIMA RMSE result I have taken from DEVELOPERS CORNER [11]. The ARIMA model got a RMSE of 72.64 that is not good prediction whereas LSTM model got RMSE of 0.43 which is very good as compared to the other model. There could be multiple reasons for this some of which can be the ARIMA model is not working good because the author is using only about five years of data to train the model. The Linear Regression and SVM RMSE I have taken from Vaishnavi Gururaj, Shriya V R and Dr. Ashwini K [12] work. We can clearly see here that LSTM is giving the best results among these models. For good prediction huge data is necessary in [10] M. Roondiwala, H. Patel and S. Varma have collected data only for 5 years if they would have collected more data, the prediction rate will be more accurate

	1 year	5 years	10 years
AAPL (RMSE)	5.6	3.2	0.43
BOFA (RMSE)	3.48	1.8	0.76
McDonald's RMSE	4.9	2.8	0.9

Table 3: RMSE for different years data

ARIMA (RMSE)	Linear Regression (RMSE)	SVM (RMSE)	LSTM (RMSE)
72.64	3.22	1.58	0.43

Table 4: RMSE Comparison with other works

2.1 Existing System

Historical stock price data is collected from reliable sources such as financial databases or APIs. The data includes features like opening price, closing price, high, low, volume, and possibly additional factors like news sentiment or economic indicators. The collected data undergoes preprocessing steps, including normalization to scale values, handling missing data, and splitting the dataset into training, validation, and test sets. Time series sequences are formatted for input into the LSTM model. A multi-layer LSTM neural network is designed, often with additional layers like dropout for regularization. The architecture includes parameters such as the number of LSTM layers, units per layer, activation functions (e.g., ReLU), and output layer configuration. The LSTM model is trained using backpropagation through time (BPTT) and an optimizer like Adam or RMSprop. Hyperparameters such as learning rate, batch size, and number of epochs are tuned to optimize model performance. The trained model is evaluated using performance metrics such as Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), or Mean Absolute Percentage Error (MAPE) on validation and test datasets. These metrics assess the accuracy of stock price predictions. The LSTM model is used to make predictions on unseen data (test set), and the results are analyzed to assess the model's effectiveness in capturing stock price trends and patterns. Visualization techniques may be used to compare predicted and actual prices. Based on the analysis, the model may be refined by adjusting hyperparameters, experimenting with different architectures (e.g., stacked LSTMs), or incorporating additional features (e.g., technical indicators, sentiment analysis). Once validated, the LSTM model can be deployed for real-time or batch predictions. Monitoring mechanisms may be implemented to track model performance over time and retrain the model periodically to adapt to changing market conditions.

2.2 Analysis

RNN:

LSTM is a special type of Recurrent Neural Network (RNN). This uses data from previous records and based on that it predicts the future. RNN can remember our previous input in their memory when a huge dataset is given Figure shows how unrolled RNN works and Figure 3 shows how Rolled RNN works. Here if we unloop the data, we can see that the first output is based on the first input and second output is based on the first and second then the third output is based on first, second and third and so on. RNN can be thought as multiple copies of the same neural network where each passing a message to the successor.

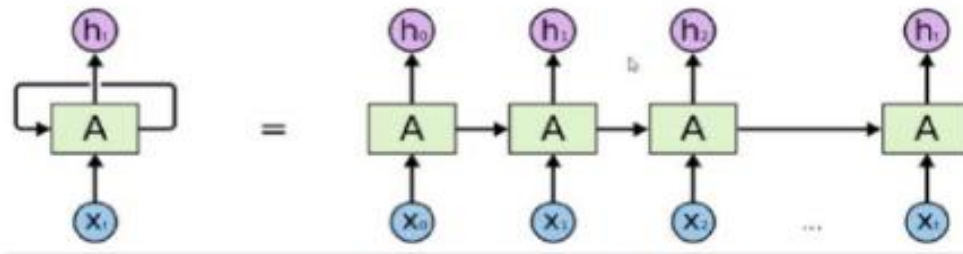


Figure 2: An unrolled RNN
(Aditi Mittal Oct 12, 2019
Understanding RNN and LSTM)

LSTM memory cell In Figure 5 we can see a single cell is updated with controlling gates which controls the previous memory as well as the input. This is how they handle vanishing and exploding gradients. Figure 6 shows clear picture of how an input gate an output gate and a forget gate are working. The cell remembers values and the gates regulate the flow of information.

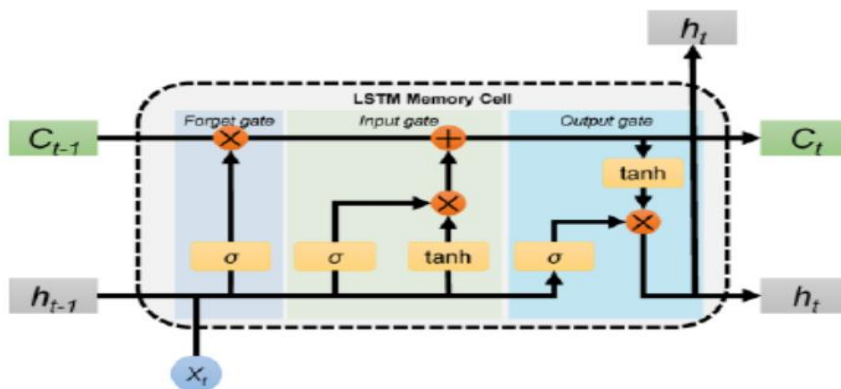


Figure 6: LSTM memory cell

2.3 Requirement Specification

2.5.1 Hardware Requirements

- ❖ Laptop or PC with minimum 8GB RAM and i5 10th Generation processor or higher.
- ❖ Core I5 processor

2.5.2 Software Requirements

- ❖ Python 3.6 or higher.
- ❖ Windows 10 OS.
- ❖ Visual studio

Chapter 3

SCOPE

SCOPE

3.1 Executive Summary

Stock market is dynamic, unpredictable due to nature of the volatile market. Predicting any stock value accurately is a huge challenge as there are so many factors to consider such as news, sentiments, economy, financial reports and much more. The strategies for investment in stock market is very complex and depends on tons of data. Profit always comes with the risk of losses. To minimize the risk of losing money and maximize the profit the techniques to predict stock value is highly useful. There are two main approaches that are being used for predicting stock values. In the first method which is Traditional Time Series method the prediction is based on the historical data of that particular stock. In this method the stock's closing price, opening price volume etc. has been used. The second method, that is qualitative, the prediction is based on factors like company profile, news articles, economy, social media, market sentiments etc. For stock market the size of the data is quite huge and random so we need models that are efficient and can deal with the complexity of this huge amount of data. The stocks data are complicated and difficult to understand due to the hidden patterns. Machine learning techniques have potential to deal with the complexity and dig to solve the multilayered complicated patterns and come up with a good prediction. The main purpose of this research is to predict future price of a particular stock. In this research I collected stock price data from Yahoo Finance to feed the data to machine learning algorithm model.

2. Related Work In the work of Kim and Han the researchers have used ANN and GA to predict future stock price. The data authors have used is from Korea stock price index (KOSPI). The sample data they have collected from KOSPI was about 10 years ranging from January 1989 to December 1998. They applied some required optimization and techniques to prepare the data to be used. They used GA to optimize ANN. For optimization they used RMSprop. They used the latest technique to perform the task, but they did not have any background knowledge in financial domain. In this paper the author did not mention how they train the model with long-time dependencies.

3.2 Business Objectives

The primary objective is to develop a robust LSTM model that accurately predicts stock price movements. By achieving high prediction accuracy, the project aims to help investors make informed decisions and maximize their investment returns.

Risk Mitigation: Provide risk management insights by identifying potential market risks and opportunities. The model should be able to detect anomalies or deviations from expected price trends, aiding in risk assessment and proactive decision-making. Develop and optimize trading strategies based on LSTM predictions. This includes identifying optimal entry and exit points, portfolio rebalancing strategies, and incorporating risk-adjusted returns to improve overall trading performance. Enable real-time decision support by creating a scalable and efficient prediction system. This allows investors and traders to access up-to-date predictions and adjust their strategies accordingly, leveraging timely market insights. Provide a competitive edge by leveraging advanced machine learning techniques like LSTM networks. The project aims to outperform traditional forecasting methods and offer more accurate and timely predictions, enhancing the competitiveness of investment strategies.

Long-Term Investment Planning: Support long-term investment planning by providing insights into stock price trends over extended periods. The model should capture both short-term fluctuations and long-term trends, aiding in strategic investment planning and asset allocation. Ultimately, the project seeks to enhance customer satisfaction by empowering investors with reliable and actionable stock price predictions. Meeting or exceeding customer expectations regarding prediction accuracy and usability is a key business objective.

3.3 Project Justification

Stock price data exhibits temporal dependencies, where past prices influence future trends. LSTM networks excel at capturing such long-range dependencies, making them well-suited for modeling sequential data like stock prices.

Stock price movements are often non-linear and influenced by multiple factors such as market sentiment, economic indicators, and geopolitical events. LSTM networks, with their ability to model complex non-linear relationships, offer advantages over traditional linear models. LSTM networks can automatically learn relevant features from raw input data, reducing the need for manual feature engineering. This capability is crucial in stock price prediction, where identifying meaningful patterns and trends is challenging.

LSTM networks are designed to handle time series data efficiently. They can learn from historical sequences of stock prices, volume, and other relevant variables, capturing nuances and patterns that impact price movements. The memory cells in LSTM networks enable them to retain important information over extended periods, mitigating the vanishing gradient problem often encountered in deep learning models trained on sequential data. LSTM models can adapt to changing market conditions and trends. They can learn from new data streams and adjust their predictions, making them valuable tools in dynamic and evolving financial markets.

Empirical studies and applications have shown that LSTM networks, when properly trained and validated, can achieve competitive prediction accuracy in stock price forecasting compared to traditional models.

Accurate stock price predictions using LSTM networks can aid investors and financial institutions in risk management strategies, portfolio optimization, and decision-making processes, leading to more informed investment decisions.

3.4 Project Description

In this project, we aim to develop a robust stock price prediction model leveraging the power of Long Short-Term Memory (LSTM) networks, a type of recurrent neural network (RNN) well-suited for sequential data analysis. The primary objective is to forecast future stock prices based on historical price data, incorporating both technical indicators and market sentiment analysis. Our LSTM model architecture is designed to capture temporal dependencies and patterns in stock price movements. It consists of multiple LSTM layers with adjustable parameters such as the number of units per layer, activation functions, dropout rates for regularization, and optimizer settings. We select relevant features that contribute to the predictive power of the model. This includes technical indicators like moving averages, Relative Strength Index (RSI), MACD (Moving Average Convergence Divergence), and other market-specific metrics. Feature engineering may involve creating lagged variables or combining multiple indicators to extract meaningful insights. The project involves a continuous improvement cycle, where we refine the LSTM model based on feedback, incorporate new data, adjust parameters, and explore advanced techniques like ensemble models or attention mechanisms to enhance predictive performance over time.

Technologies Used:

- Python: For backend logic, image processing, and model integration.
- OpenCV: Library for image and video processing tasks.
- Keras or TensorFlow: Deep learning frameworks for emotion recognition model integration.
- HTML, CSS, JavaScript: Frontend development for user interface and interaction.
- Visual Studio: Integrated development environment (IDE) for Python development and Flask application setup.

3.5 Risk Assessment

The LSTM model may not accurately capture complex market dynamics, leading to inaccurate predictions. Conduct rigorous testing and validation on historical data, use ensemble techniques to improve stability, and monitor model performance regularly. Insufficient or low-quality historical data can lead to biased or unreliable predictions. Ensure data integrity and reliability, consider alternative data sources or augmentation techniques, and handle missing or noisy data effectively. Sudden market shifts or unpredictable events can affect stock prices, challenging the model's ability to adapt. Implement risk management strategies, such as incorporating volatility measures into the model, using robust optimization techniques, and diversifying the portfolio. The model may overfit to historical data, resulting in poor generalization to new market conditions. Apply regularization techniques (e.g., dropout, L2 regularization), use cross-validation to assess generalization performance, and consider simpler model architectures to avoid overfitting. Complex LSTM models may lack interpretability, making it challenging to understand the rationale behind predictions. Use model explainability techniques (e.g., SHAP values, feature importance analysis), visualize model predictions and insights, and document model assumptions and limitations.

3.6 Constraints

Potential constraints may arise from data quality issues such as missing values, outliers, or inconsistent data formats. LSTM models, especially with deep architectures, can be computationally intensive and require substantial training time. Constraints related to hardware resources, memory limitations, or scalability issues may affect the model's performance and deployment feasibility. Risks associated with overfitting, where the model memorizes training data patterns but fails to generalize to new data, need to be addressed. Constraints on dataset size, diversity, and representativeness can contribute to overfitting challenges. Optimizing LSTM model hyperparameters such as learning rate, batch size, number of layers, and units per layer is critical for model performance. Constraints in conducting extensive hyperparameter tuning experiments due to time or resource limitations can impact model optimization. Constraints related to modeling complex market behaviors and adapting to changing market conditions need to be considered. Compliance with regulatory requirements, such as data privacy (e.g., GDPR, CCPA) and financial regulations, imposes constraints on data handling, model development, and deployment processes. Risks related to biased predictions, unfair outcomes, or unintended consequences of using predictive models in financial decision-making require ethical considerations. Constraints on ensuring fairness, transparency, and accountability in model development and usage are paramount.

Chapter 4

PROBLEM STATEMENT

PROBLEM STATEMENT

The objective of this project is to develop an accurate and robust model for predicting stock prices based on historical data using Long Short-Term Memory (LSTM) networks.

The main challenges addressed in this endeavor include:

- ❖ Stock prices exhibit complex temporal dependencies and non-linear patterns, making them challenging to predict using traditional methods. LSTM networks are well-suited for capturing long-term dependencies in sequential data, making them a suitable choice for this task.
- ❖ Financial markets are influenced by a multitude of factors, including economic indicators, news events, and investor sentiments. Incorporating diverse data sources and handling data variability are essential to improve prediction accuracy.
- ❖ Stock markets are inherently volatile and subject to sudden fluctuations due to external factors such as geopolitical events, regulatory changes, or market sentiment shifts. Developing a robust model capable of adapting to changing market conditions is crucial for reliable predictions.
- ❖ In addition to prediction accuracy, the interpretability of the model's predictions is vital for informed decision-making. Understanding the factors driving the model's predictions and identifying significant features contribute to building trust in the predictive model.
- ❖ By addressing these challenges, the project aims to deliver a predictive model that provides valuable insights into future stock price movements, aiding investors, financial analysts, and decision-makers in making informed investment decisions.

Chapter 5

PROPOSED SYSTEM

PROPOSED SYSTEM

As represented getting the historical data from market is mandatory step. Then we will have to extract the feature, which is required for data analysis, then divide it as testing and training data, training the algorithm to predict the price and the final step it to visualize the data. Figure 4 represents the Architecture of the proposed system.

- Step 1: Data Collection.
- Step 2: Data Preprocessing after getting the historic data from the market for a particular share and read the close price
- Step 3: Divide the data
- Step 4: Do a feature scaling on the data so that the data values will range between 0 and 1.
- Step 5: Building the RNN (Recurrent neural network) for data set and Initialize the RNN by using sequential regressor.
- Step 6: Building the LSTM model
- Step 7: Train the model by providing batch size and epochs. • Step 8: Making the predictions and visualizing the results using plotting techniques

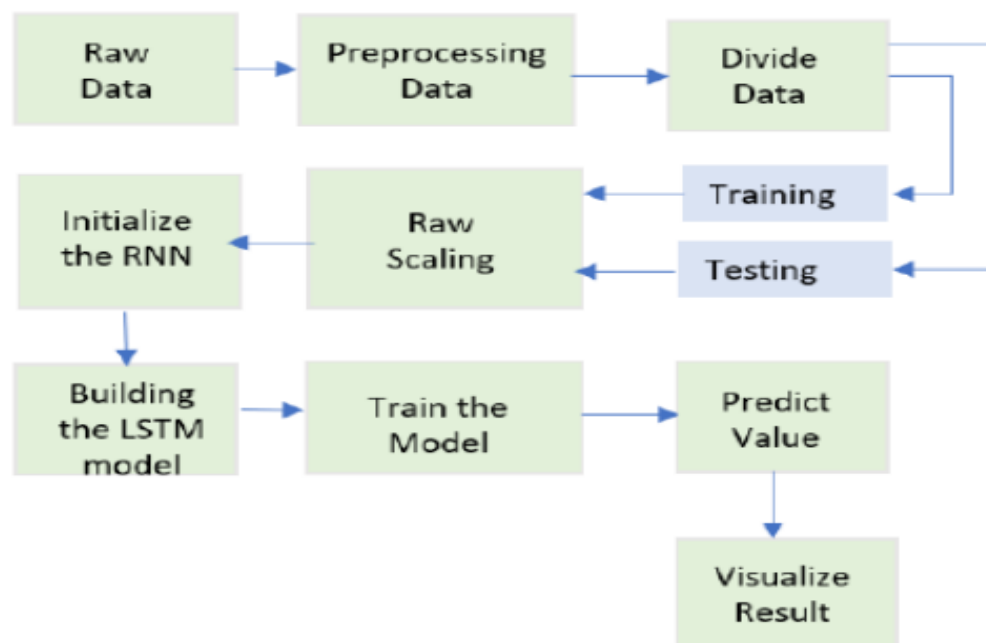


Figure 4: Proposed system

CHAPTER 6

SYSTEM DESIGN

DESIGN

6.1 Architectural Block Diagram

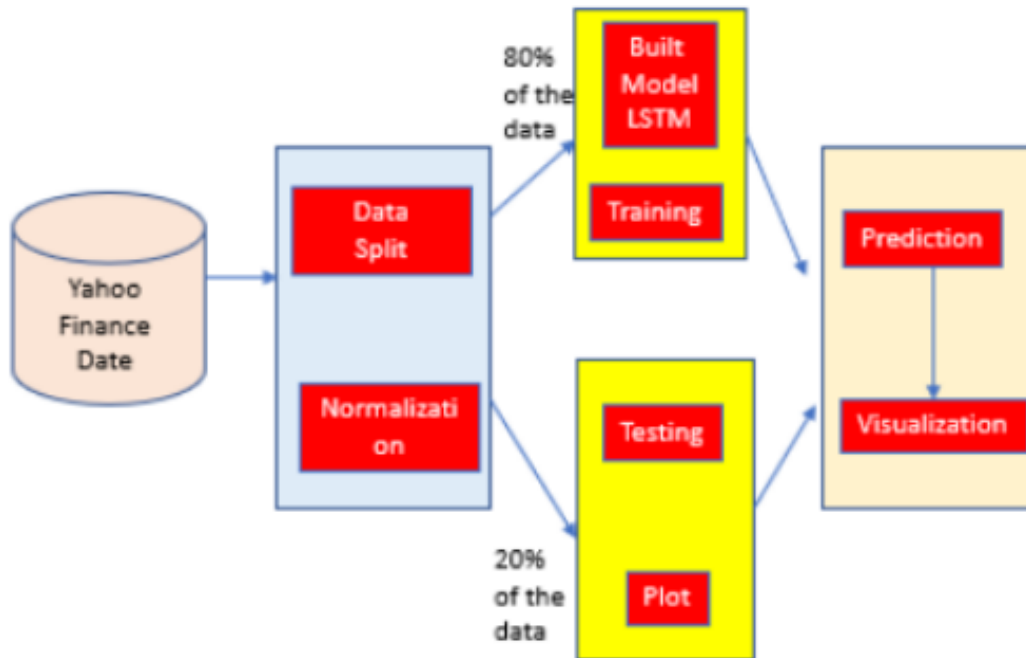


Fig 6.1 architectural diagram of the proposed system

To build the LSTM model we will need to import Keras library and packages. Keras is basically tensor flow high-level API for building and training models. Here we are going to import some libraries. Sequential – It is basically linear stack of layers through which we can create a sequential layer by passing the list by it. Dense - It is the regular deeply connected neural network layer. It is most commonly used layer to change the dimension of the output. It represents a metrics vector multiplication so the value which are trainable parameter get updated during back propagation. Next, we are going to initialize RNN. For time series problems like this we are going to use regression model. The first step in this is to read the data which is a sequential data, and we are going to assign this to regressor. After this we are going to move to the next step that is the most important stage training the model. In this stage the data we collected is going to fed to the model and trained for prediction.

6.2 Use Case Diagram

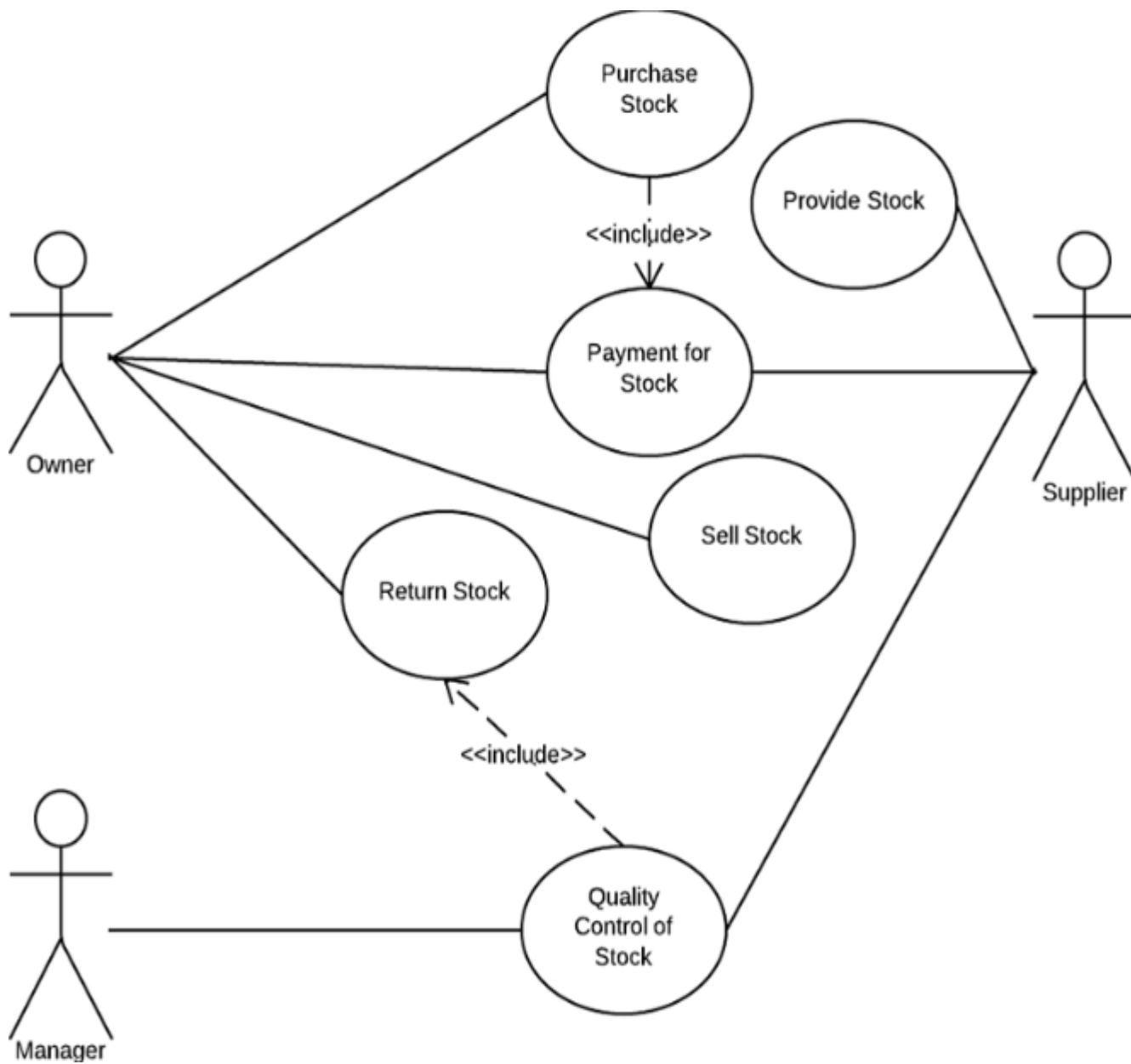
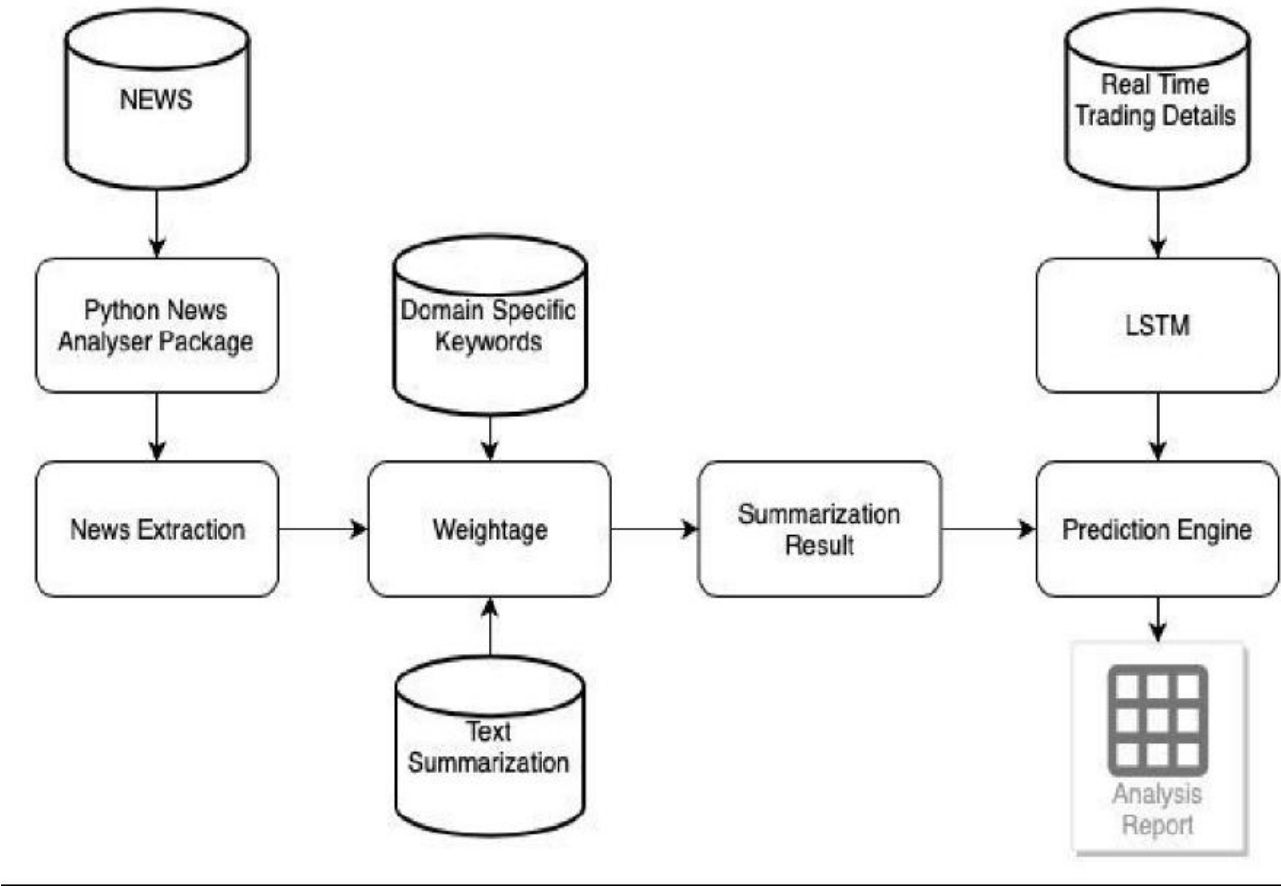


Fig 6.2 Use Case Diagram

6.3 Data Flow Diagram



6.4Flow Chart

Flow Chart: The following flowchart shows the flow of data and loops of the program.

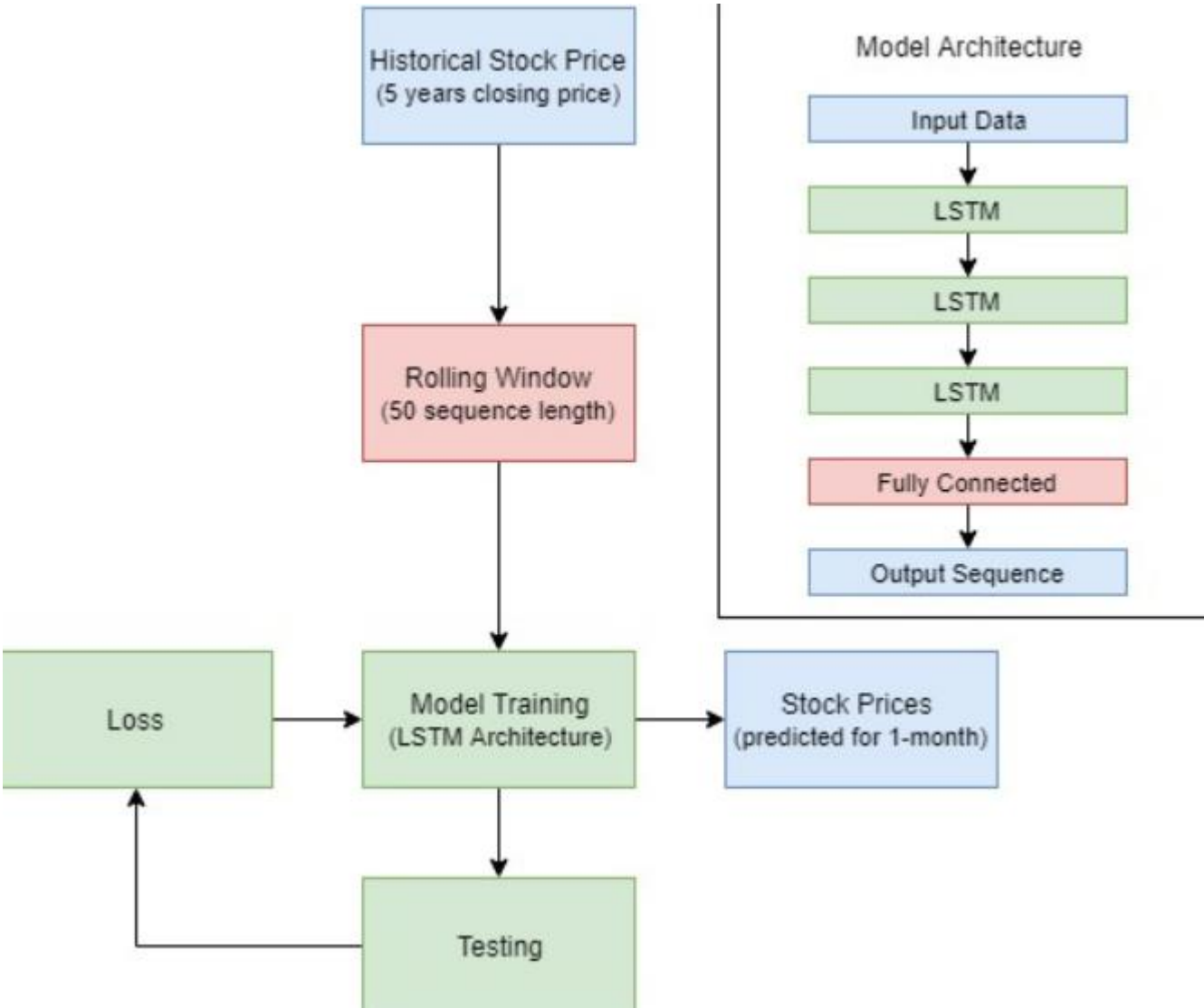


Fig 6.6. Flow Chart

Chapter 7

IMPLEMENTATION

7.1 Implementation Results

To build the LSTM model we will need to import Keras library and packages. Keras is basically tensor flow high-level API for building and training models. Here we are going to import some libraries. Sequential – It is basically linear stack of layers through which we can create a sequential layer by passing the list by it. Dense - It is the regular deeply connected neural network layer. It is most commonly used layer to change the dimension of the output. It represents a metrics vector multiplication so the value which are trainable parameter get updated during back propagation. Next, we are going to initialize RNN. For time series problems like this we are going to use regression model. The first step in this is to read the data which is a sequential data, and we are going to assign this to regressor. After this we are going to move to the next step that is the most important stage training the model. In this stage the data we collected is going to be fed to the model and trained for prediction.

Stock Market Predictor

Enter Stock Symbol

GOOG

Stock Data

Date	Open	High	Low	Close	Adj Close	Volume
2013-01-17 00:00:00	17.8757	17.9238	17.7091	17.7166	17.7166	88,791,570
2013-01-18 00:00:00	17.6927	17.7527	17.4678	17.547	17.547	129,555,794
2013-01-22 00:00:00	17.5507	17.5677	17.3231	17.5061	17.5061	152,264,594
2013-01-23 00:00:00	18.331	18.6551	18.3261	18.4683	18.4683	237,249,950
2013-01-24 00:00:00	18.4618	18.8501	18.4436	18.7848	18.7848	135,815,168
2013-01-25 00:00:00	18.6992	18.8912	18.6862	18.7714	18.7714	89,369,729
2013-01-28 00:00:00	18.7238	18.8195	18.6274	18.6982	18.6982	65,327,951
2013-01-29 00:00:00	18.599	18.8531	18.5938	18.7716	18.7716	70,145,942
2013-01-30 00:00:00	18.7731	18.9527	18.7525	18.7754	18.7754	69,579,828
2013-01-31 00:00:00	18.6927	18.8698	18.6862	18.8217	18.8217	65,613,015

Price vs MA50



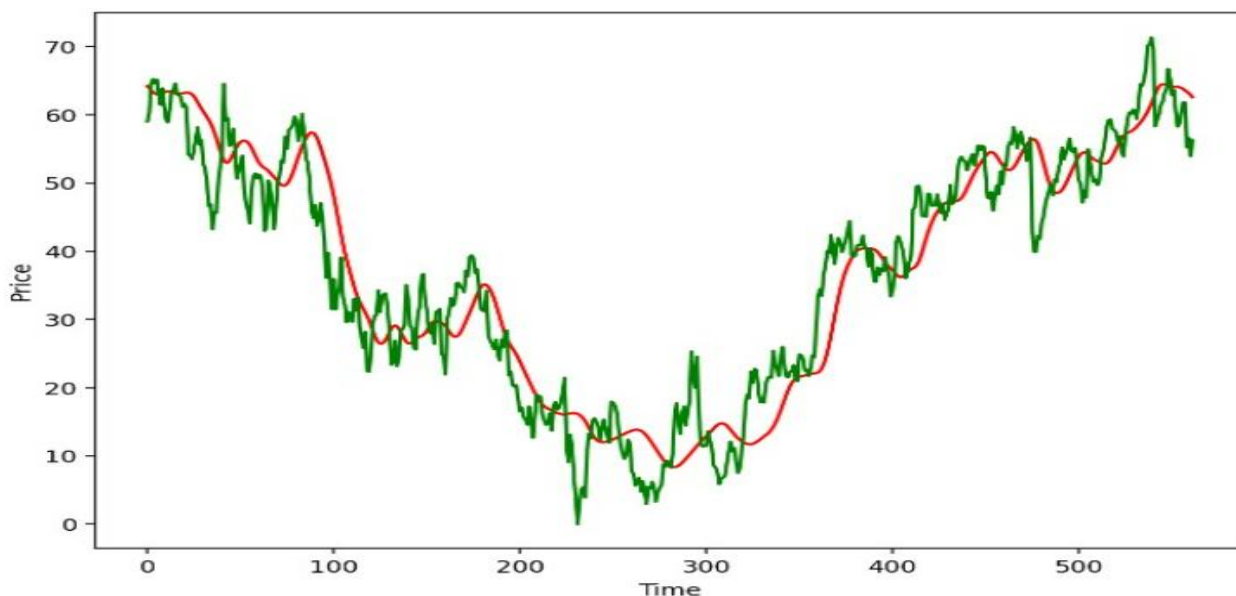
Price vs MA50 vs MA100



Price vs MA100 vs MA200



Original Price vs Predicted Price



This LSTM model is composed of a sequential input layer followed by three LSTM layers and a dense layer. In the model we are also using dropout. The dropout is a regularization technique for reducing overfitting, so it drops out units in neural networks.

7.2 Result Analysis

After training the result of the prediction we need to show the result. The result could be shown in many ways but the most effective way to show the result in a concise view is through a Graph. In Graph we can show the comparison between the result of both actual values and prediction value using the data that were collected and processed. In these graphs (Graphs 1, 2, and 3), the blue line (Train) is showing the data used in training, the red line (Val) is the actual value of the data and orange line (Prediction) is the graph for predicted data. We can see actual and predicted values are going in similar direction. In all the graphs we see that both lines are very close to each other. Any model cannot predict stock value accurately for many reasons such as if there is any positive or negative news come from that company the stock prices will have an impact. Furthermore, if some sudden changes in market happens, we cannot predict that beforehand. There could be multiple reasons the stock price can fluctuate aberrant. In this project, I am trying to feed the pattern of the stock price for a particular company and based on that I am creating a model that will be trained with the historical data and based on the pattern it will apply an algorithm and will predict the future value. Thus, we cannot be completely sure in predicting the stock price. The analysis shows that to predict the value of such kind of activity is not a straight task because there are many factors and problems that should be considered before making a prediction. To understand this better, have also generated values in table format. This is easier to read and understand for any specific values. In the table, we can easily figure out what the actual value of the specified date is and what the model is predicting for the same date.

Chapter 8

MINIMUM SOFTWARE AND HARDWARE REQUIREMENTS

MINIMUM SOFTWARE AND HARDWARE REQUIREMENTS

8.1 Hardware Requirements

- ❖ Laptop or PC with minimum 8GB RAM and i5 10th Generation processor or higher.
- ❖ Edge browser

8.2 Software Requirements

- ❖ Python 3.6 or higher.
- ❖ Windows 10 OS.
- ❖ Visual Studio

8.3 Libraries

- ❖ NumPy stands for Numerical Python and it can be used to perform a wide variety of mathematical operations on arrays. It adds powerful data structures to Python that guarantee efficient calculations with arrays and matrices and it supplies an enormous library of high-level mathematical functions that operate on these arrays and matrices.
- ❖ Keras is a minimalist Python library for deep learning that can run on top of Theano or TensorFlow. It was developed to make implementing deep learning models as fast and easy as possible for research and development.

Chapter 9

TIME LINE CHART

TIME LINE CHART

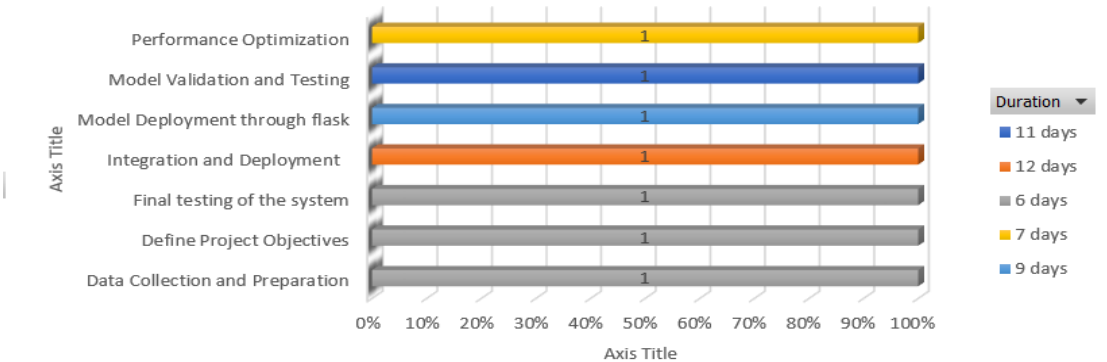
9.1 Summary of Work done

Gathered historical stock price data from financial sources and prepared it by normalizing, handling missing values, and splitting into training, validation, and test sets.

9.2 Time Line Chart

Table 9.2 Time Line Chart

Event	Start Date	Duration	End Date
Define Project Objectives	27/01/2024	6 days	03/02/2024
Data Collection and Preparation	04/02/2024	6 days	10/02/2024
Model Deployment through flask	11/02/2024	9 days	10/02/2024
Model Validation and Testing	21/02/2024	11 days	02/03/2024
Integration and Deployment	03/03/3024	12 days	14/03/2024
Performance Optimization	15/03/2024	7 days	21/03/2024
Final testing of the system	24/03/2024	6 days	30/03/2024



Chapter 10

TESTCASES

10.1 Testcases

Software Testing is an activity that aims at evaluating an attribute or a capability of a system and determining that it meets its required results. This chapter identifies the testcases for our proposed system.

Test Case ID	Test Case Execution Steps	Expected Results	Tester	Test Result
TC1	Gather historical stock price data for a specific stock or index. Preprocess the data by normalizing, handling missing values, and splitting into training, validation, and test sets.	The data is prepared and formatted correctly for model training and evaluation.	Abhishek	Pass
TC2	Train an LSTM model using the prepared historical data. Optimize hyperparameters (e.g., learning rate, batch size) for model convergence and accuracy.	The LSTM model converges during training, and optimization improves its predictive capabilities.	Abhishek	Pass
TC3	Validate the trained LSTM model by evaluating predictions on a validation dataset. Use metrics like Mean Absolute Error (MAE) or Root Mean Squared Error (RMSE) to assess prediction accuracy.	The model achieves an accuracy rate of at least 80% on the validation dataset.	Abhishek	Pass
TC4	Test model predictions on unseen test data and compare against actual stock prices.	The model's predictions closely match actual stock prices, indicating robustness and generalization.	Abhishek	Pass
TC5	Assess model performance under different market conditions (e.g., bull market, bear market, high volatility) to evaluate robustness.	The model demonstrates stable performance across varying market scenarios, showing resilience to market fluctuations.	Abhishek	Pass
TC6	Implement a live prediction test using real-time stock price data to validate model responsiveness and accuracy in dynamic market scenarios.	The model provides timely and accurate predictions when applied to real-time market data.	Abhishek	Pass

Chapter 11

CONCLUSION

CONCLUSION

In conclusion we can say that the popularity of stock market trading is growing rapidly which is encouraging researchers to find out new methods for the prediction using new techniques. The forecasting techniques is not only helpful to researchers, but it also helps investors or any person dealing with the stock market. To help predict stock indices or forecasting model with good accuracy is required. Overall, we conclude that Stock prediction using Machine learning can increase the prediction accuracy. Through historical data we can find the pattern of the stock and how the stocks movement has been over years and learn through them. A model can learn the trends with huge amount of data provided and train themselves to predict the movement of the stock price. However, it is also important to understand the parameters that influence the prediction in order to develop an efficient product with accuracy of results. We can conclude that historical data can be used as powerful tool that allows us to forecast the movement more accurately. However, our method might not generate the accurate result in this case for the number of reasons mentioned such as any negative or positive news or sudden change in market etc. Machine learning is the best way to recognize patterns and predict results rather than apply the analysis manually on large sets of data. There are some drawbacks that need to consider while applying the algorithm to predict the result. Nobody should blindly rely on these models and invest since there are number of reasons that the prediction can go in other direction.

Chapter 12

FUTURE SCOPE

11.1 Future Scope

The model used in my project has shown very promising result. From the various results shown through graphs and tables in the research can confirm that my model is capable of tracing the evolution of closing prices. For our future work we will try to find the best sets for about data length and number of training epochs that better suit our assets and maximize our prediction accuracy.

Chapter 13

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Chapter 14

ACKNOWLEDGEMENT

ACKNOWLEDGEMENT

The success and final outcome of required a lot of guidance and assistance from many people and we are extremely privileged to have got this all along the completion of our project. All that we have done is only due to such supervision and assistance and we would not forget to thank them.

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We are thankful to and fortunate enough to get constant encouragement support and guidance from all Teaching staffs of Information Technology. Also, we would like to extend our warm thanks to our family members and well-wishers for always supporting and encouraging us.

Yours sincerely,

Abhishek Bharat Kadam

Chapter 15

APPENDIX A: CODE SAMPLE

APPENDIX A: CODE SAMPLE

Keras.ipynb

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import yfinance as yf

start = '2012-01-01'
end = '2022-12-21'
stock = 'GOOG'

data = yf.download(stock, start, end)

data.reset_index(inplace=True)

data

ma_100_days = data.Close.rolling(100).mean()

plt.figure(figsize=(8,6))
plt.plot(ma_100_days, 'r')
plt.plot(data.Close, 'g')
plt.show()

ma_200_days = data.Close.rolling(200).mean()

plt.figure(figsize=(8,6))
plt.plot(ma_100_days, 'r')
plt.plot(ma_200_days, 'b')
plt.plot(data.Close, 'g')
plt.show()

data.dropna(inplace=True)

data_train = pd.DataFrame(data.Close[0: int(len(data)*0.80)])
data_test = pd.DataFrame(data.Close[int(len(data)*0.80): len(data)])

data_train.shape[0]

data_test.shape[0]
```

```

from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler(feature_range=(0,1))

data_train_scale = scaler.fit_transform(data_train)

x = []
y = []

for i in range(100, data_train_scale.shape[0]):
    x.append(data_train_scale[i-100:i])
    y.append(data_train_scale[i,0])

x, y = np.array(x), np.array(y)

from keras.layers import Dense, Dropout, LSTM
from keras.models import Sequential

model = Sequential()
model.add(LSTM(units = 50, activation = 'relu', return_sequences = True,
               input_shape = ((x.shape[1],1))))
model.add(Dropout(0.2))

model.add(LSTM(units = 60, activation='relu', return_sequences = True))
model.add(Dropout(0.3))

model.add(LSTM(units = 80, activation = 'relu', return_sequences = True))
model.add(Dropout(0.4))

model.add(LSTM(units = 120, activation = 'relu'))
model.add(Dropout(0.5))

model.add(Dense(units =1))

model.compile(optimizer = 'adam', loss = 'mean_squared_error')

model.fit(x,y, epochs = 50, batch_size =32, verbose =1)

model.summary()

pas_100_days = data_train.tail(100)

data_test = pd.concat([pas_100_days, data_test], ignore_index=True)

data_test_scale = scaler.fit_transform(data_test)

x = []
y = []

```



```

for i in range(100, data_test_scale.shape[0]):
    x.append(data_test_scale[i-100:i])
    y.append(data_test_scale[i,0])
x, y = np.array(x), np.array(y)

y_predict = model.predict(x)

scale = 1/scaler.scale_

y_predict = y_predict*scale

y = y*scale

plt.figure(figsize=(10,8))
plt.plot(y_predict, 'r', label = 'Predicted Price')
plt.plot(y, 'g', label = 'Original Price')
plt.xlabel('Time')
plt.ylabel('Price')
plt.legend()
plt.show()

model.save('Stock Predictions Model.keras')

```

App.py

```

import numpy as np
import pandas as pd
import yfinance as yf
from keras.models import load_model
import streamlit as st
import matplotlib.pyplot as plt

model = load_model(r'C:\Users\sejal\OneDrive\Documents\Project\Updated stock market
prediction\Stock Predictions Model.keras')

st.header('Stock Market Predictor')

stock =st.text_input('Enter Stock Symbnol', 'GOOG')
start = '2023-01-01'
end = '2024-06-21'

data = yf.download(stock, start ,end)

```

```

st.subheader('Stock Data')
st.write(data)

data_train = pd.DataFrame(data.Close[0: int(len(data)*0.80)])
data_test = pd.DataFrame(data.Close[int(len(data)*0.80): len(data)])

from sklearn.preprocessing import MinMaxScaler
scaler = MinMaxScaler(feature_range=(0,1))

pas_100_days = data_train.tail(100)
data_test = pd.concat([pas_100_days, data_test], ignore_index=True)
data_test_scale = scaler.fit_transform(data_test)

st.subheader('Price vs MA50')
ma_50_days = data.Close.rolling(50).mean()
fig1 = plt.figure(figsize=(8,6))
plt.plot(ma_50_days, 'r')
plt.plot(data.Close, 'g')
plt.show()
st.pyplot(fig1)

st.subheader('Price vs MA50 vs MA100')
ma_100_days = data.Close.rolling(100).mean()
fig2 = plt.figure(figsize=(8,6))
plt.plot(ma_50_days, 'r')
plt.plot(ma_100_days, 'b')
plt.plot(data.Close, 'g')
plt.show()
st.pyplot(fig2)

st.subheader('Price vs MA100 vs MA200')
ma_200_days = data.Close.rolling(200).mean()
fig3 = plt.figure(figsize=(8,6))
plt.plot(ma_100_days, 'r')
plt.plot(ma_200_days, 'b')
plt.plot(data.Close, 'g')
plt.show()
st.pyplot(fig3)

x = []
y = []

for i in range(100, data_test_scale.shape[0]):
    x.append(data_test_scale[i-100:i])
    y.append(data_test_scale[i,0])

x,y = np.array(x), np.array(y)

```

```
predict = model.predict(x)

scale = 1/scaler.scale_

predict = predict * scale
y = y * scale

st.subheader('Original Price vs Predicted Price')
fig4 = plt.figure(figsize=(8,6))
plt.plot(predict, 'r', label='Original Price')
plt.plot(y, 'g', label = 'Predicted Price')
plt.xlabel('Time')
plt.ylabel('Price')
plt.show()
st.pyplot(fig4)
```